

# AI & Machine Learning for Civil Engineers

From engineering data to prediction, explainability and uncertainty, in eight hands-on sessions.

**08** Live sessions**90** min Per session**04** weeks Course duration

 **Runs 100% on Google Colab**  
Nothing to install, works even on a Chromebook

 **Built for beginners**  
No prior Python or machine learning required

 **Live code-along on Civil Engineering datasets**  
Materials · Buildings & Energy · Transport · Environment · Geotechnics

 **Hands-on notebooks, cross-domain assignments & solutions**  
Plus certificate & recordings (till 2 weeks after the course)

## Course outline

**01 Python for Civil Engineers**  
AI/ML basics, Colab, NumPy & pandas with engineering data

**02 From Raw Data to ML-Ready**  
Explore, clean, visualize & split engineering data

**03 Your First Prediction Model**  
Linear regression, error metrics & cross-validation

**04 Trees & Ensemble Models**  
Decision Trees, Random Forest & boosting for nonlinear problems

 **Designed for Civil Engineers across disciplines** — Materials, Structures, Geotechnics, Transportation, Water & Environment.

 **Early Bird offer till 12 October**

Regular fees apply after 12 Oct

### WHO SHOULD JOIN

B.Tech & M.Tech students / PhD scholars / Faculty /  
Practising engineers / Complete beginners to AI & ML



### INSTRUCTOR

**Dr. Abdul Waris**

Research Associate

Universität Siegen (University of Siegen)

Former Prime Minister's Research Fellow

Ph.D. in Geotechnical Engineering, IIT Hyderabad

 Research Excellence Award 2026, IIT Hyderabad

 Invited talks at Oregon State University (USA) and a  
Faculty Development Programme at D.Y. Patil (Pune, India)

 Resource person, five-day ML workshop at NIT Srinagar

**05 Classification for Engineering Decisions**  
Logistic regression, confusion matrix, precision & recall

**06 Artificial Neural Networks**  
Architecture, activations, training & overfitting

**07 Explainable AI & Engineering Judgement**  
Feature importance, SHAP, and engineering interpretation

**08 From Prediction to Probability**  
Monte Carlo, ML surrogates & probability of failure

STARTS · LIMITED SEATS

**Saturday,  
17 October 2026**

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